

Truly Innovative **TECH**

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Tech*

Photon-counting image sensor

Gigajot Technology presents photon-counting image sensor family that is suitable for scientific, medical, defence, industrial and other high-performance applications. The device has a sub-

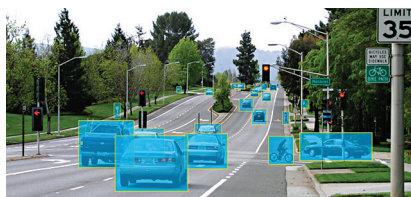


electron read noise capability of < 0.2 electron RMS and can detect dark current lower than 0.1 electron/pixel/sec at room temperature, thus giving a high resolution. Accurate photon-counting is performed up to 100s and 1,000s of photoelectrons. Developed using stacked backside illuminated CMOS process, the image sensor offers high quantum efficiency and excellent pixel isolation. Other features include excellent low-light imaging performance, on-chip temperature sensor, and high intra-scene dynamic range.

Gigajot Technology
www.gigajot.tech

Predictive lidar

Autonomous Vehicle (AV) technology company Insight LiDAR has announced that its product, Insight 1600, is the first lidar with a combined high resolution

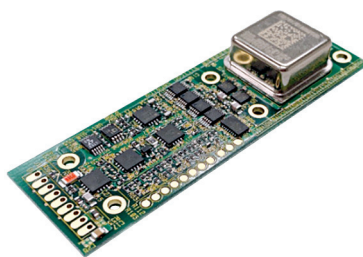


and velocity detection to enable pedestrian gesture recognition. The new capability can be used by AV perception teams to quickly and accurately detect and predict subtle actions of pedestrian movements. It conveys important safety capability that had eluded AV developers until now. The device's ultra-high resolution and instantaneous low-velocity detection with every pixel plays a critical role in making vehicle decisions, especially regarding other vehicle movement.

Insight LiDAR
www.insightlidar.com

High-stability MEMS gyro

Designed for use in severe environments including downhole drilling operations, platform stabilisation, and autonomous vehicle control, Silicon Sensing Systems presents its ultra-high stability gyroscope, the CRS39A. It incorporates micro

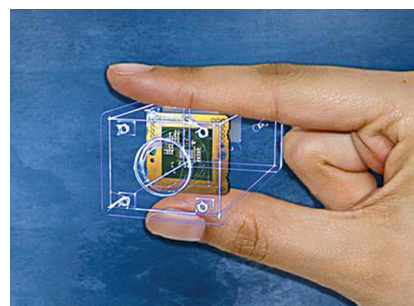


electro-mechanical systems and electronics, including new drive electronics and improvements to the sensor head. These have delivered significant performance enhancements in critical areas such as bias instability, angle random walk, noise and vibration tolerance whilst multiple on-board temperature sensors allow CRS39A performance to be tuned to suit each operating environment. The device is a cost-effective and rugged alternative to fibre-optic and dynamically tuned gyros.

Silicon Sensing Systems
www.siliconsensing.com

Embedded vision system

Vision Components presents a small embedded vision system, VC picoSmart, that comes fully integrated in a single board and is hardly bigger than an image sensor module. Only 22mm × 23.5mm



in size, this board camera contains all the components necessary for image processing: 1-megapixel CMOS sensor with global shutter, FPGA module, high-end FPU processor, and memory. An interface board can be connected via the integrated FPC connector, which makes VC picoSmart the perfect base for rapid development of cost-effective, application-specific, and compact vision sensor. The company's VCRT operating system enables real-time image processing for object recognition, position control, barcode reading, as well as web edge and level control.

Vision Components GmbH
www.vision-components.com

Solution for external portable SSDs

Silicon Motion Technology Corporation, a designer of NAND flash controllers for solid-state storage devices, has launched the SM2320 single-chip high-performance, lower-power and cost-effective solution for external portable SSDs. It is designed with integrated hardware and firmware as well as high-level security features, which meet the needs of game